

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2 : Scenario-Based

**COURSE NAME:** Cloud Computing and Big Data Analytics      **COURSE CODE:** CSA1506

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### COURSE OUTCOME COVERED

**CO2:** *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)  
Question Count: 5

**Total Marks: 25**

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### Code of Conduct

I \_M.Mary Sharlin ( 192511330 )\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: \_M.Mary Sharlin\_

### Assessment Tasks

#### Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

## **ANSWERS:**

1.

### **Introduction**

A media company needs to support millions of concurrent users with minimum buffering. A scalable cloud architecture with caching, load balancing, and auto-scaling is suitable.

### **Architecture**

Users → DNS → Load Balancer → Auto-Scaling Servers → Cache/CDN → Cloud Storage

- Cloud Storage: Stores videos in different formats and qualities.
- CDN: Caches videos near users to reduce latency and buffering.
- Load Balancer: Distributes requests among multiple servers.
- Auto-scaling: Automatically increases or decreases servers according to traffic.
- Database: Stores user information, subscriptions, and video metadata.

### **Scalability**

- Use horizontal scaling by adding more application servers.
- Deploy resources across multiple availability zones.
- Use CDN caching for frequently accessed videos.

### **Load Balancing**

The load balancer distributes incoming users among healthy servers. This prevents a single server from becoming overloaded.

### **Auto-Scaling**

During peak hours, new instances are automatically created. During low traffic, unnecessary instances are removed.

## **Cost Optimization**

- Use pay-as-you-go cloud resources.
- Use auto-scaling to avoid unused servers.
- Store frequently accessed data in faster storage and old videos in cheaper storage.
- Use CDN caching to reduce bandwidth and server load.

## **Conclusion**

Using CDN, caching, load balancing, auto-scaling, and cloud storage provides smooth streaming with high scalability and optimized cost.

2.

## **Introduction**

The fintech startup uses containers across multiple cloud providers. Container orchestration can provide consistent deployment and management.

## **Proposed Architecture**

Applications → Containers → Container Orchestrator → Cloud Provider A / Cloud Provider B

- Use Docker containers to package applications consistently.
- Use Kubernetes for container orchestration.
- Deploy applications across multiple cloud providers.
- Use CI/CD pipelines for automated deployment.

## **Container Orchestration**

### **Kubernetes manages:**

- Container deployment
- Scheduling
- Health checks
- Scaling
- Service management
- Automatic recovery

## **Service Discovery**

Services can communicate using Kubernetes service names and internal DNS. This avoids depending on fixed IP addresses.

## **Scaling**

Horizontal Pod Autoscaling can increase or decrease application containers based on CPU, memory, or traffic.

### **Benefits of Multi-Cloud**

- Avoids dependence on one cloud provider.
- Improves availability.
- Provides disaster recovery.
- Allows organizations to select suitable services from different providers.

### **Risks**

- Increased complexity.
- Difficult monitoring and management.
- Different cloud APIs and services.
- Higher operational cost.
- Security and compliance challenges.

### **Conclusion**

Kubernetes + containers + CI/CD + multi-cloud deployment provides portability, scalability, and improved availability.

3.

### **Introduction**

The e-learning platform experienced data loss because of accidental deletion. A cloud-native backup and disaster recovery strategy is required.

### **Proposed Solution**

Application → Primary Database → Backup → Replication → Disaster Recovery Environment

### **Backup**

- Take regular automated backups.
- Store backups separately from the primary system.
- Use scheduled full and incremental backups.

### **Versioning**

Enable versioning for cloud storage so accidentally deleted or modified files can be restored to previous versions.

### **Replication**

Replicate important data to another availability zone or region. If the primary system fails, the replicated copy can be used.

## **RTO**

**Recovery Time Objective (RTO)** is the maximum acceptable time required to restore services.

- Use automated recovery.
- Maintain ready-to-use backup infrastructure.
- Automate deployment and restoration.

## **RPO**

**Recovery Point Objective (RPO)** defines how much recent data loss is acceptable.

- Frequent backups and continuous replication reduce RPO.
- Critical databases should be replicated frequently.

## **Business Continuity Steps**

1. Detect the failure.
2. Stop affected services if necessary.
3. Identify the latest valid backup.
4. Restore data.
5. Switch users to the recovery environment.
6. Verify application functionality.
7. Resume normal operations.

## **Conclusion**

**Backup, versioning, replication, low RTO, and low RPO** ensure data protection and business continuity.

4.

## **Introduction**

The government agency needs to keep sensitive data on-premises while using cloud resources for non-sensitive data.

## **Architecture**

Users → Identity Management → Secure Connection → On-Premises + Public Cloud

- **On-premises:** Sensitive data and applications requiring strict compliance.

- **Public cloud:** Non-sensitive applications and scalable workloads.
- **VPN/Private Connection:** Securely connects both environments.

## Secure Communication

- Use encrypted VPN or dedicated private connectivity.
- Use firewalls and security groups.
- Control traffic between on-premises and cloud networks.

## Identity Management

Implement centralized identity and access management:

- Role-Based Access Control (RBAC)
- Multi-factor authentication
- Least-privilege access
- Centralized user management

## Encryption

- Encrypt data **at rest** using cloud encryption.
- Encrypt data **in transit** using secure communication protocols.
- Manage encryption keys securely.

## Challenges

- Complex infrastructure management.
- Different security policies.
- Network latency.
- Compliance requirements.
- Higher monitoring and maintenance effort.
- Integration between on-premises and cloud systems.

## Conclusion

A **secure hybrid cloud using private connectivity, IAM, encryption, and access control** provides flexibility while protecting sensitive government data.

5.

## Introduction

A SaaS provider must continue operating even if one cloud region fails. A multi-region architecture provides high availability and fault tolerance.

## Architecture

Users → Global DNS/Load Balancer → Region A / Region B → Application Servers → Databases

- Deploy applications in at least two regions.
- Use load balancing to distribute traffic.
- Keep application instances in multiple availability zones.

## **Data Replication**

Data should be replicated between regions.

- Use synchronous replication when very low data loss is required.
- Use asynchronous replication when performance and distance are important.
- Maintain regular backups.

## **Failover**

### **If Region A fails:**

1. Monitoring detects the failure.
2. Global load balancer/DNS identifies the healthy region.
3. Traffic is redirected to Region B.
4. Users continue accessing the application.
5. Failed region is restored later.

## **Fault Tolerance**

- Use multiple application instances.
- Avoid single points of failure.
- Use redundant databases and networks.
- Deploy resources across availability zones.

## **Monitoring and Alerting**

### **Monitor:**

- CPU and memory usage
- Application errors
- Network latency
- Database health
- Server availability

- Traffic levels

Set alerts for abnormal conditions and automatically notify administrators.

## Conclusion

A multi-region architecture with replication, automatic failover, redundancy, monitoring, and alerts ensures high availability and continuous SaaS operation.

## Scenario-Based Rubric

| Criteria                         | Weightage | Level 4 - Exemplary                                         | Level 3 - Proficient                 | Level 2 - Developing          | Level 1 - Beginning              |
|----------------------------------|-----------|-------------------------------------------------------------|--------------------------------------|-------------------------------|----------------------------------|
| Understanding of Scenario        | 20%       | Demonstrates complete understanding of the scenario context | Good understanding with minor gaps   | Partial understanding         | Fails to understand the scenario |
| Identification of Risks / Issues | 20%       | Identifies all major issues correctly                       | Identifies most issues               | Limited issue identification  | Misses major issues              |
| Application of Concepts          | 25%       | Applies virtualization concepts effectively                 | Applies concepts with minor mistakes | Partially correct application | Weak application                 |
| Decision Justification           | 20%       | Decisions are well justified and technically strong         | Reasonable justification             | Weak justification            | No useful justification          |
| Clarity                          | 15%       | Clear and well-organized response                           | Generally clear                      | Somewhat unclear              | Poorly presented                 |